

Leaning against inflation experiences

by **Stefan Nagel**

Discussion by Oliver Pfäuti (UT Austin)

NBER New Developments in Long-Term Asset Management

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The paper on one slide

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(Hanson and Stein, 2015; Giglio and Kelly, 2018)
- ▶ secular interest rate changes accompanied by secular patterns in forecast errors
(Cieslak, 2018)

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+ outlook & some minor comments

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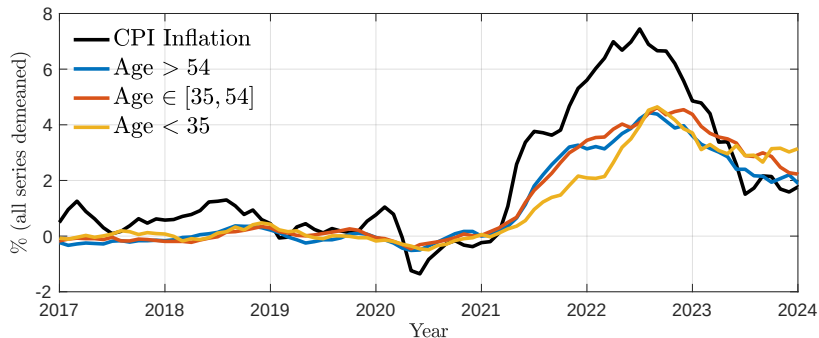
- ⇒ the updating gain is decreasing with age
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What about the recent inflation surge?

- ⇒ look at Michigan Survey of Consumers by age (demeaned by 2015-2019 averages)

The role of experiences: short-run expectations

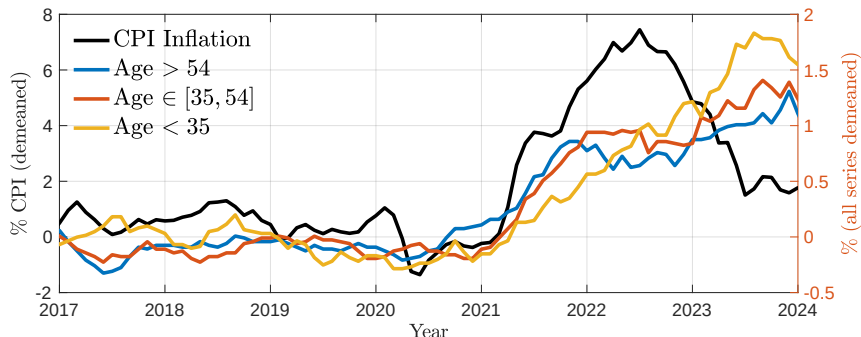
1-year ahead expectations:



⇒ older households increased their expectations earlier & more strongly in 2021

The role of experiences: medium-to-long-run expectations

5-10 years ahead expectations (right axis):

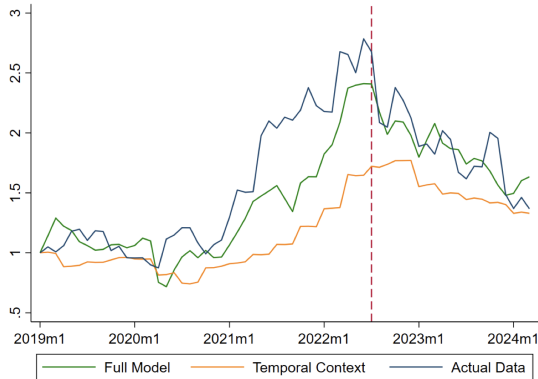


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Gennaioli, Leva, Schoenle, and Shleifer (2024)

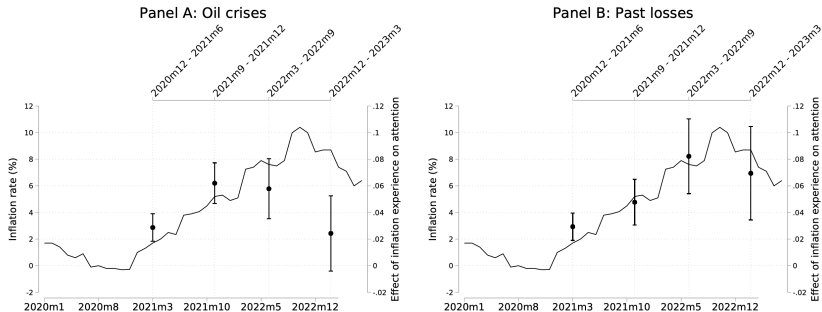
- ▶ document similar patterns for the NY Fed Survey of Consumer Expectations
- ▶ **selective memory recall**: when observing high inflation, people who **experienced** high inflation in the past recall those memories and increase expectations sharply

Figure 3. Nowcasting: Re-Anchoring of Inflation Expectations



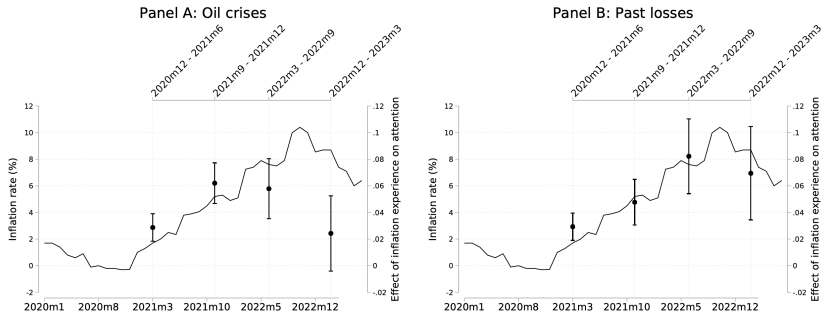
Link, Peichl, Pfäuti, Roth, and Wohlfart (2025)

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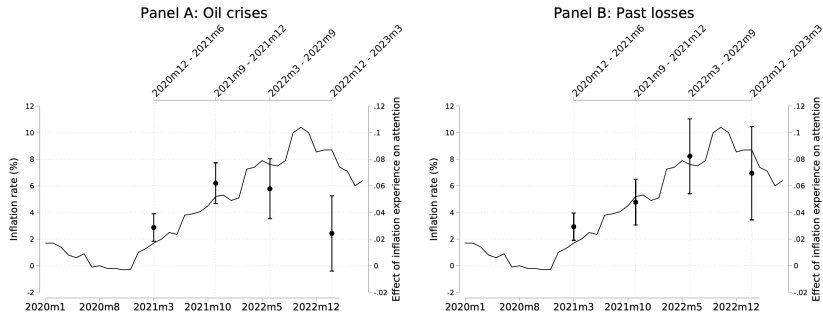
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- ▶ selective memory recall embedded in an otherwise-rational inattention framework can account for these patterns

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 - ⇒ matters for inflation outlook (more later)

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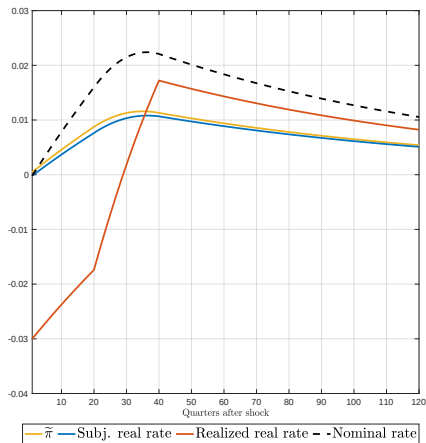
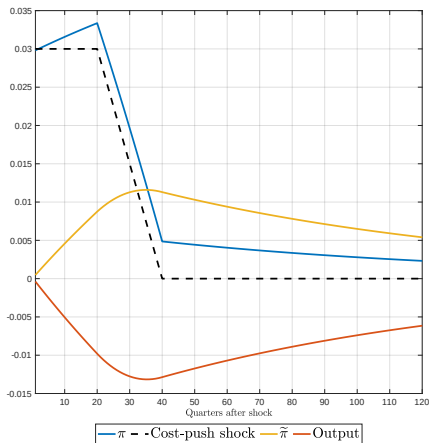
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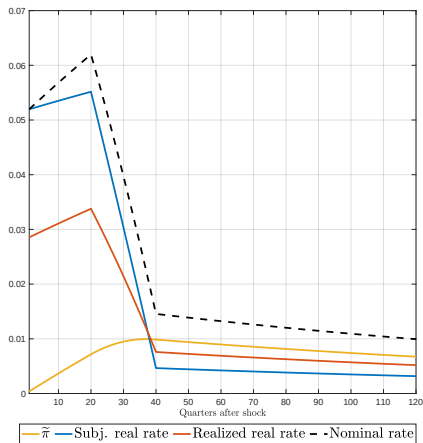
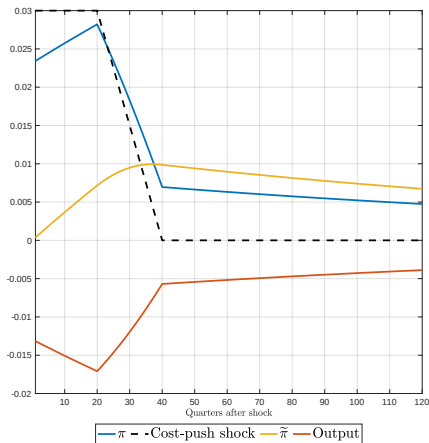
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What does monetary policy lean against?

The dynamics of real rates differ substantially across these two scenarios

1. ex-post real rates fall initially and only slowly increase when MP leans against $\tilde{\pi}$, then stay persistently high
2. ex-post real rates increase immediately and strongly, but then come down relatively quickly under “standard” formulation

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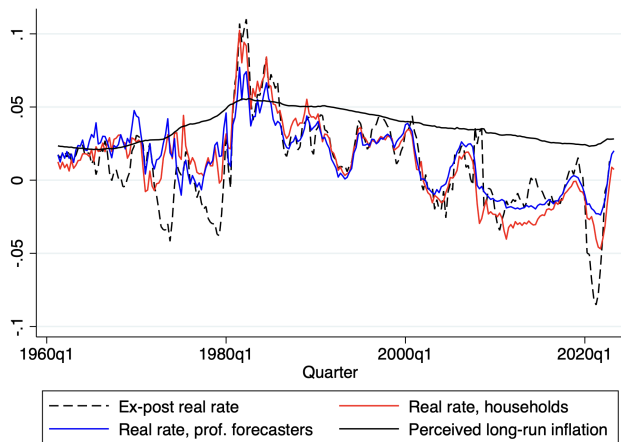
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Which one is it?

Real rate dynamics and long-run inflation

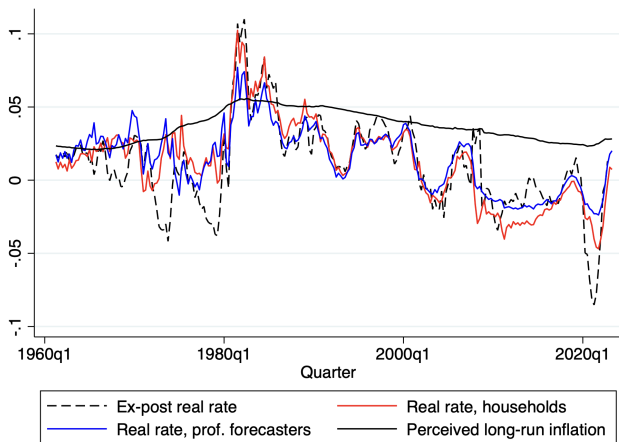
(A) Real interest rates at one-year horizon



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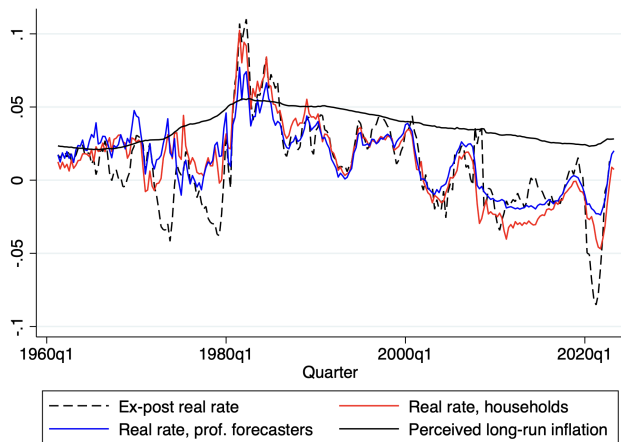


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⇒ but what *should* monetary policy do?

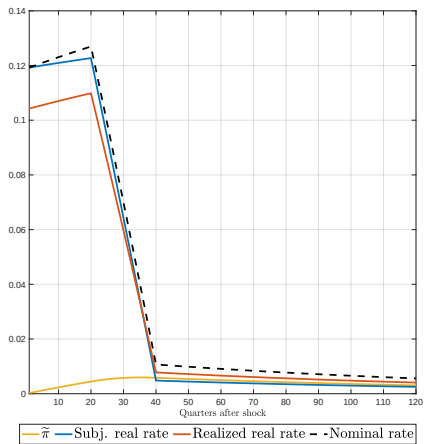
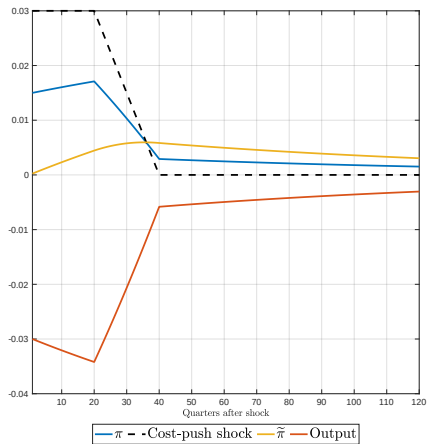
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Using (ex-post) central bank loss:

$$W = \sum_{t=0}^{\infty} \frac{1}{2} \beta^t (\pi_t^2 + \lambda x_t^2)$$

Next: solve for optimal monetary policy under discretion

The dynamics under optimal monetary policy



Optimal MP: $\Rightarrow$ lean strongly and swiftly against inflation (not $\tilde{\pi}$)

Implications for monetary policy

“Hawkish” optimal policy prescription aligns well with recent learning literature:

- ▶ Gáti (2023): if long-run expectations unanchor, optimal MP responds aggressively
- ▶ Erceg, Lindé, and Trabandt (2024): “looking through” supply shocks not necessarily a good policy when shocks are large
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Outlook:

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 - ↪ more inflationary pressure even absent new shocks
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⇒ risk of another inflationary episode higher than absent these channels

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- ▶ How important is the assumption that in the NK model only long-run inflation expectations matter? (See, e.g., Werning (2022))
- ▶ What about factors driving long-run real rates that are not captured by r^* ? (AI, green transition, geopolitical changes...)

Conclusion

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- ▶ Paper provides a lot of food for thought:
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Thank you!

References

- BULLARD, J. B., A. GRIMAUD, I. SALLE, AND G. VERMANDEL (2025): “Soft landing and inflation scares,” .
- CIESLAK, A. (2018): “Short-rate expectations and unexpected returns in treasury bonds,” *The Review of Financial Studies*, 31, 3265–3306.
- ERCEG, C., J. LINDÉ, AND M. TRABANDT (2024): “Monetary policy and inflation scares,” *Unpublished, International Monetary Fund, Washington, DC*.
- GÁTI, L. (2023): “Monetary policy & anchored expectations—An endogenous gain learning model,” *Journal of Monetary Economics*.
- GENNAIOLI, N., M. LEVA, R. SCHOENLE, AND A. SHLEIFER (2024): “How inflation expectations de-anchor: The role of selective memory cues,” *Working paper*.
- GIGLIO, S. AND B. KELLY (2018): “Excess volatility: Beyond discount rates,” *The Quarterly Journal of Economics*, 133, 71–127.
- HANSON, S. G. AND J. C. STEIN (2015): “Monetary policy and long-term real rates,” *Journal of Financial Economics*, 115, 429–448.
- LINK, S., A. PEICHL, O. PFÄUTI, C. ROTH, AND J. WOHLFART (2025): “Attention to the Macroeconomy,” *Working Paper*.
- MALMENDIER, U. AND S. NAGEL (2016): “Learning from inflation experiences,” *The Quarterly Journal of Economics*, 131, 53–87.
- NAGEL, S. (2024): “Leaning against inflation experiences,” *Unpublished manuscript, University of Chicago*.
- PFÄUTI, O. (2023): “The inflation attention threshold and inflation surges,” *Working paper*.
- WERNING, I. (2022): “Expectations and the Rate of Inflation,” Tech. rep., National Bureau of Economic Research.